

Automatic Detection of Parkinson's Disease from Gait Signals Using Time-Series Feature Extraction and Machine Learning

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Abstract: Parkinson's disease (PD) is a progressive neurodegenerative disorder that affects motor control and gait patterns. Early and reliable detection of PD remains challenging using traditional clinical assessment methods, which often rely on subjective evaluation by neurologists. Recent advances in wearable sensors and biomedical signal analysis have enabled the development of automated computational approaches for detecting Parkinsonian gait abnormalities.

This project investigates the effectiveness of machine learning models for detecting Parkinson's disease from gait signals recorded in the PhysioNet GaitPDB dataset, which contains vertical ground reaction force measurements collected during walking from 93 Parkinson's patients and 73 healthy control subjects. Time-series features were extracted using the TSFresh framework, which automatically computes a large set of statistical and frequency-domain descriptors from time-series signals.

Several classification algorithms were evaluated, including Random Forest, Extra Trees, CatBoost, and Support Vector Machines (SVM) with linear and radial basis function kernels. The models were trained and evaluated using five-fold cross-validation, and performance was assessed using accuracy, sensitivity, specificity, and area under the ROC curve (AUC).

Among the evaluated models, CatBoost achieved the highest classification accuracy (0.80) using left-foot features, while ExtraTrees achieved the highest AUC (0.874) when combining signals from both feet. The results demonstrate that feature-based machine learning models can effectively detect Parkinson's disease from gait signals, with tree-based ensemble models consistently outperforming SVM-based approaches. These findings highlight the potential of time-series feature extraction combined with ensemble learning for automated neurological disorder detection.

1. Introduction

Parkinson's disease (PD) is one of the most common neurodegenerative disorders, affecting more than 10 million people worldwide [1]. The disease primarily affects the dopaminergic neurons in the substantia nigra, leading to motor symptoms such as tremor, rigidity, bradykinesia, and postural instability. Among these symptoms, gait abnormalities are one of the earliest and most observable indicators of Parkinson's disease progression [2].

Clinical diagnosis of PD typically relies on neurological examination and patient history. However, these methods may be subjective and can lead to delayed diagnosis, especially during early stages of the disease. Consequently, there has been growing interest in automated computational methods that analyze gait signals using wearable sensors and machine learning techniques [3].

Ground reaction force signals collected from pressure sensors embedded in shoes provide detailed information about walking dynamics and foot-ground interactions. Previous studies have shown that Parkinson's patients exhibit distinct gait patterns, including reduced stride length, irregular step timing, and asymmetric foot movement [4]. These abnormalities can be detected using time-series analysis and machine learning models.

Time-series feature extraction frameworks such as TSFresh (Time Series Feature Extraction based on Scalable Hypothesis Tests) allow automatic computation of hundreds of statistical descriptors from raw signals. These features capture both temporal and frequency-domain characteristics of gait signals, enabling machine learning algorithms to learn discriminative patterns for disease detection [5].

This project aims to evaluate the effectiveness of machine learning models in detecting Parkinson's disease using gait signals from the PhysioNet GaitPDB dataset. The study focuses on three main aspects:

1. Evaluating the classification performance of multiple machine learning models.
2. Investigating whether combining signals from both feet improves detection performance.
3. Comparing the robustness of tree-based ensemble models and SVM-based models.

The key research question addressed in this study is:

Can machine learning models trained on TSFresh features effectively detect Parkinson's disease from gait signals?

To answer this question, systematic experiments were conducted using five-fold cross-validation, and models were evaluated using accuracy, sensitivity, specificity, and AUC metrics.

2. Methodology

2.1 Dataset

The experiments in this project were conducted using the GaitPDB dataset from PhysioNet, a publicly available biomedical dataset containing ground reaction force signals collected from individuals during walking [6].

The dataset contains recordings from:

- 93 Parkinson's disease patients
- 73 healthy control subjects

Each participant walked at a natural pace for approximately two minutes, while vertical ground reaction forces (vGRF) were recorded using pressure sensors embedded in footwear.

These time-series signals represent the force applied by the foot to the ground during walking, providing detailed insight into gait dynamics.

Diagram 1: Example of raw gait time-series signal

(See Presentation Slide 14)

The dataset provides measurements for both feet, enabling experiments with three input configurations:

- Left foot signals (L)
- Right foot signals (R)
- Combined signals from both feet (LR)

This design allows investigation of whether bilateral gait information improves classification performance.

2.2 Time-Series Feature Extraction

To convert raw gait signals into machine-learning-compatible features, the TSFresh framework was used [5]. TSFresh automatically computes a large number of statistical descriptors from time-series data, including:

- Mean, variance, and standard deviation
- Autocorrelation coefficients
- Entropy measures
- Frequency-domain statistics
- Energy and peak characteristics

These features capture both temporal dynamics and statistical properties of gait signals. Feature extraction was performed separately for:

1. Left-foot signals
2. Right-foot signals
3. Combined signals from both feet

This allowed analysis of whether gait asymmetry contributes to improved Parkinson's detection.

Diagram 2: Feature extraction pipeline using TSFresh

(See Presentation Slide 19)

2.3 Machine Learning Models

Several classification algorithms were evaluated in this study:

- Random Forest
- Extra Trees
- CatBoost
- Support Vector Machine (Linear kernel)
- Support Vector Machine (RBF kernel)

Tree-based ensemble models such as Random Forest and Extra Trees are known for their ability to model nonlinear relationships and interactions between features [7]. CatBoost is a gradient boosting algorithm that provides strong performance on structured datasets and is designed to reduce overfitting [8]. Support Vector Machines were included as baseline models to compare traditional kernel-based approaches with ensemble learning methods.

2.4 Training Strategy

To ensure robust evaluation, five-fold cross-validation was applied.

The dataset was divided into five equal subsets:

- Four folds used for training
- One fold used for testing

This process was repeated five times so that each subset served as the test set once.

Model performance was evaluated using the following metrics:

- Accuracy
- Sensitivity (Recall for PD class)
- Specificity (True negative rate)
- Area Under the ROC Curve (AUC)

3. Results

3.1 Model Performance Comparison

The performance of the evaluated models across different input configurations is summarized in Table 1.

Diagram 3: Overall model performance comparison

(See Presentation Slide 45)

The results show that tree-based ensemble models consistently outperform SVM-based models.

Key findings include:

- CatBoost achieved the highest classification accuracy (0.80)
- ExtraTrees achieved the highest AUC (0.874)
- Random Forest produced stable performance across all input configurations

3.2 AUC Trend Across Input Types

To analyze the effect of different input configurations, AUC performance was evaluated for left, right, and combined signals.

Diagram 4: AUC trend across input types

(See Presentation Slide 51)

The results show that:

- AUC values generally improve when combining left and right signals (LR)
- Tree-based models maintain stable performance across inputs
- SVM-RBF shows high sensitivity to input choice

3.3 Accuracy Analysis

Accuracy trends across input types further confirm the advantage of combining signals from both feet.

Diagram 5: Accuracy trend across input types

(See Presentation Slide 53)

Tree-based models such as CatBoost and ExtraTrees maintain stable accuracy, while SVM models show larger fluctuations.

3.4 Confusion Matrix Analysis

To better understand classification behavior, confusion matrices were analyzed for the best performing models.

Diagram 6: Confusion matrices for CatBoost
(See Presentation Slide 55)

The results show that CatBoost achieves:

- High true-positive detection of PD cases
- Balanced classification performance across classes

Diagram 7: Confusion matrices for Random Forest
(See Presentation Slide 57)

The Random Forest model also demonstrates stable classification performance, although slightly lower accuracy compared to CatBoost.

4. Discussion and Limitations

The experimental results demonstrate that machine learning models can effectively detect Parkinson's disease from gait signals using TSFresh features.

However, several limitations should be considered.

First, the dataset used in this project contains a relatively small number of subjects, which may limit the generalizability of the results. Larger datasets would enable more robust evaluation.

Second, TSFresh generates a large number of features, which may introduce redundant information. Feature selection or dimensionality reduction techniques such as PCA or mutual information filtering could improve model efficiency.

Third, this project focused on classical machine learning models, while modern deep learning architectures such as recurrent neural networks or state-space models could potentially capture temporal dependencies more effectively.

Despite these limitations, the study provides a strong baseline framework for Parkinson's disease detection using gait signals.

5. Conclusion

This project investigated the use of machine learning techniques for automatic detection of Parkinson's disease using gait signals from the PhysioNet GaitPDB dataset.

Time-series features were extracted using the TSFresh framework, and several classification models were evaluated using five-fold cross-validation.

Among the evaluated methods:

- CatBoost achieved the highest classification accuracy (0.80)
- ExtraTrees achieved the highest AUC performance (0.874)

Tree-based ensemble models consistently outperformed Support Vector Machines, highlighting their effectiveness in handling high-dimensional time-series features.

The results demonstrate that gait signals contain meaningful patterns that can be used for automated Parkinson's disease detection. Future work could explore deep learning models and larger datasets to further improve performance and enable clinical deployment.

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